

Taylor R. Brown

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Education

Ph.D. Statistics, The University of Virginia, 2018.
M.S. Statistics, The University of Connecticut, 2013.
B.A. Mathematics & Economics, The University of Connecticut, 2010.

Employment

Assistant Professor of Statistics, General Faculty, Department of Statistics, University of Virginia 2021–.
Lecturer of Statistics, Department of Statistics, University of Virginia 2018-2021.

Articles and Papers

A Case-Study of Sample-Based Bayesian Forecasting Algorithms, 2022.
[arXiv:2208.02968](https://arxiv.org/abs/2208.02968) [[stat.AP](#)]

The Most Difference in Means: A Statistic for Null and Near-Zero Results, 2022.
[arXiv:2201.01239](https://arxiv.org/abs/2201.01239) [[stat.ME](#)]

A Short Introduction to PF: A c++ Library for Particle Filtering, 2020.
[The Journal of Open Source Software](#)

Approximating Posterior Predictive Distributions by Averaging Output From Many Particle Filters, 2020.
[arXiv:2006.15396](https://arxiv.org/abs/2006.15396) [[stat.ME](#)]

PF: A C++ Library for Fast Particle Filtering, 2020.
[arXiv:2001.10451](https://arxiv.org/abs/2001.10451) [[stat.CO](#)]

A Pseudo-Marginal Metropolis-Hastings Algorithm for Estimating Generalized Linear Models in the Presence of Missing Data, 2019.
[arXiv:1907.09090](https://arxiv.org/abs/1907.09090) [[stat.ME](#)]

A Markov-Switching Factor Stochastic Volatility Model, 2018.
[arXiv:1903.01841v1](https://arxiv.org/abs/1903.01841v1) [[stat.AP](#)]

Brown, Taylor. *Factor Stochastic Volatility Models for Portfolio Construction*. University of Virginia, Department of Statistics, PHD (Doctor of Philosophy), 2018, <https://doi.org/10.18130/V3ZW18R9V>

Books

An Introduction to R and Python For Data Analysis: A Side By Side Approach, 2021
[CRC Press Taylor & Francis Group](#)

Public Software

pf: a C++ library for fast particle filtering.
<https://github.com/tbrown122387/pf>

pfR: Interface to the C++ Library pf
<https://cran.r-project.org/web/packages/pfr/index.html>

gradeR: helps grade R script assignment submissions!
<https://cran.r-project.org/web/packages/gradeR/index.html>

cPseudoMaRg: Constructs a Correlated Pseudo-Marginal Sampler
<https://cran.r-project.org/web/packages/cPseudoMaRg/index.html>

ssme: a c++ static library for state space model forecasting and estimation.
<https://github.com/tbrown122387/ssme>

Teaching

University of Virginia

DS 6040: Bayesian Machine Learning

STAT 8120: Topics in Statistics (particle filtering)

STAT 7510: Advanced Topics in Statistical Inference

STAT 7620: Advanced Topics in Probability

STAT 7200: Introduction to Advanced Probability

STAT 6440: Introduction to Bayesian Methods

STAT 6021: Linear Models for Data Science

STAT 5430: Statistical Computing with SAS/Python and R

STAT 5170: Applied Time Series

STAT 4993: Independent Study

STAT 4170: Financial Time Series and Forecasting

STAT 4160: Experimental Design

STAT 3250: Data Analysis With Python

STAT 3080: From Data To Knowledge

STAT 3120: Introduction to Mathematical Statistics

STAT 2120: Introduction to Statistical Analysis

STAT 1602: Introduction to Data Science with Python

STAT 1601: Introduction to Data Science with Python

University of Connecticut

STAT 1000 Introduction to Statistics

Departmental Service

University of Virginia

Committees

Computing Committee (Chair from Spring 2025–Present)

Undergraduate Committee (Member from Fall 2021–Present)

Graduate Committee (Fall 2020–Spring 2022)

Lecturer Hiring Committee (Summer 2022)

Student Support and Advising

Undergraduate Major Advisor (Fall 2021–Present)

Computer Skills Confirmation Test (formerly the Computer Language Exam) (2018–Present)

PhD Dissertation Committees

Priyanjali Bukke's PhD Committee (2026–2027)

Tyler Ashoff's PhD Committee (2025–2026)

Tianyuan Zhou's PhD Committee (2022–2023)

Professional Service

Journal Reviewer

Journal of the Royal Statistical Society: Series A (JRSS-A)

Journal of Open Source Software (JOSS)

Honors and awards

- 2021: Learning Technology Incubator Grant
- 2016: Summer Fellowship U.Va. Department of Statistics
- 2015: Summer Fellowship U.Va. Department of Statistics
- 2014: Summer Fellowship U.Va. Department of Statistics
- 2014: U.Va. Jefferson Trust Big Data Fellowship pre-proposal award
- 2014: U.Va. Graduate Statistics Seminar Committee Co-Chair
- 2013: UConn Ross D. MacKinnon Graduate Fellowship

Memberships

- American Statistical Association
- International Society for Bayesian Analysis
- Institute of Mathematical Statistics